

Alexander Ferrari Miller

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TECHNICAL / QA / SOFTWARE

Professional Summary

Software quality assurance and development professional with commercial experience spanning software testing, test-tool development, defect analysis, network support, database testing, web technology, and cryptographic software. Earlier work includes responsibility for the charting subsystem of Lotus 1-2-3 Release 3 and QA work involving pen-computing products, merchant-management systems, SQL databases, and legal-office networks. Current independent work includes software architecture, Ada development, AI usability research, computational astronomy, web development, digital publishing, and other technical projects.

Technical Skills

Quality Assurance: Software testing, test design, test-tool development, defect analysis, bug tracking, white-box testing, black-box testing, boundary-condition testing, regression testing, usability analysis, reproducibility analysis

Software & Data: Ada, SQL, Microsoft Access, spreadsheet modeling, algorithm design, cryptographic software, database analysis

Web & Systems: HTML, web development, Jekyll, Laravel, Server-Side Includes (SSI), Linux/Unix environments, network troubleshooting, web administration

Technical Communication: Technical writing, documentation, independent research, computational astronomy, requirements and architecture development

PROFESSIONAL TECHNICAL EXPERIENCE

Software Engineer, Quality Assurance — Charting

Lotus Development Corporation — Cambridge, MA

- One of four QA engineers responsible for the charting subsystem of Lotus 1-2-3 Release 3.
- Tested charting functionality including numerical XY charts and boundary conditions.
- Designed and developed the **XY Star** visual test tool to investigate serious failures involving XY data points outside defined chart boundaries.
- Built 14 nested mathematically generated star plots whose scale, translation, and rotation could be systematically varied with predictable visual results.
- Provided developers with reproducible visual evidence that assisted in identifying and resolving charting defects.
- Developed a separate spreadsheet cell-comparison test tool using a heuristic definition of cell equality incorporating values, source contents, and formatting.
- Selected by Lotus to travel from Massachusetts to California to demonstrate Lotus 1-2-3 Release 3 to customers, using the XY Star charts as a central part of the presentation.

Software Design, Development & Testing

Sturdy Software Incorporated

- Collaboratively designed cryptographic algorithms for a working software encryption system.
- Designed reversible transformations exhibiting strong avalanche behavior in forward and inverse operations.
- Personally wrote and tested every line of code in the working encryption implementation.
- Developed reduced-scale 7-bit, 6-bit, and 5-bit implementations to expose internal behavior and test boundary conditions.
- Identified cyclic behavior in a 4-bit implementation and experimentally demonstrated that a single-bit perturbation could break the cycle.
- Participated in conceptual design and planning for proposed error-control and data-compression product families.

Senior Quality Assurance Engineer

Notable Technologies

- Performed QA for multiple pen-enabled and pen-centric applications, including a crossword game, data-compression software, and a communications application.
- Investigated performance problems associated with the constrained underlying pen-computing environment.
- Developed a bug-tracking database.
- Advocated for white-box testing and closer integration between development and QA.

Software Quality Assurance Contractor

Slate Corporation

- Tested software applications developed for pen-enabled computing environments.
- Identified lack of file-save capability as a major obstacle to effective testing.
- Advocated for preservation of test outputs so QA could compare behavior and provide concrete defect evidence.
- Documented reproducible test steps despite limitations on preserving product output.

Network Engineer

Lynch, Marks & Associates — Berkeley, CA

- Set up standardized workstations and troubleshoot computer and network problems in-office and at client locations.
- Worked with Microsoft Word and WordPerfect scripts used by law firms to generate pleadings, proofs of service, and other routine documents.
- Developed a Lotus Notes application for a client to track telemarketing calls and follow-up activity.

Software Quality Assurance Contractor

Bank of America — Concord, CA

- Performed black-box testing of a merchant-management system.
- Coordinated the systems-integration team into the testing effort.

Software Quality Assurance Contractor

Visa — Foster City, CA

- Tested Visa's merchant-management system.
- Examined the system's SQL database as part of the testing effort.

Intern

Internet Alfredo — San Francisco, CA

- Worked on dynamic website development using Server-Side Includes (SSI).
- Explored assembling dynamic web content using SSI in ways analogous to functionality later commonly implemented with server-side scripting.
- Developed Bannerclub.com, an independent project for coordinating banner advertisements for clients.

Website Designer

Pearlsom — Los Angeles, CA — January 2014–Present

- Designed, developed, and maintained web content and site organization.
- Applied technical troubleshooting and computer skills to long-term website development.
- Worked with photos, video, mailing lists, email identity, social-media integration, text chat, and webcam/video communication.

CURRENT TECHNICAL & INDEPENDENT PROJECTS

Sturdy Cryptographic Algorithm Reconstruction

Reconstructed the earlier Sturdy cryptographic algorithm in Ada from surviving knowledge of the original design. Current work focuses on preservation, compilation, test-harness development, behavioral verification, and eventual cryptographic validation.

ChannelDial

Developing an Ada-based Unix/Linux directory for collecting, organizing, searching, and verifying communications channels, with a structured data model, planned Laravel web interface, and tutorial documentation developed alongside the software.

Horse-Racing Expert System

Designed a rule-based contrarian handicapping strategy and historical-testing methodology using BRIS racing data and Microsoft Access, emphasizing profit/loss behavior across betting odds rather than winner-picking accuracy alone.

Pearlsom

Developing a centralized personal web platform integrating media, mailing lists, email identity, social-media linking, chat, and video communication while preserving users' existing investment in outside social networks.

Fusslor

Developing a related social-web platform with working capabilities for profiles, media publishing, messaging, mailing lists, digital content, and connections to external social identities.

Elephant

Designed a preservation-oriented AI architecture intended to retain significant conversational information with provenance while orchestrating external tools and services. Related Elephant Book work explores AI-assisted notebook organization and retrieval.

The Illuminated Bible

Designed and prototyped a multilingual publishing system that can illuminate significant words in an English Biblical text with Hebrew, Aramaic, Syriac, or Koine Greek source-language information, transliteration, and translation.

Star Almanack

Developing an observer-first computational astronomy and publishing system combining astronomical calculations, regression testing, observational guidance, graphical presentation, and technical writing.

ChatGPT Usability Study

Conducting an independent structured software-quality study of recurring persistence, interface, and behavioral problems encountered during sustained real-world use, evaluated for severity, user impact, reproducibility, and effects.

LunarFAX

Designed a device-based fax application concept using an accumulating page allowance, with a proposed Lunar Facts educational astronomy campaign as one application.

Afghanistan Saffron Business

Developing a business project involving Afghan saffron; detailed technical scope, business model, and implementation status are being documented as the project develops.

GitHelm

Developing a software project whose architecture, purpose, implementation status, and technical details are being documented as development proceeds.

The Heather Papers

A prose project preserving and organizing extended discussions about artificial intelligence, its behavior, implications, and the human experience of working with it.

AlexanderFerrariMiller.com — Reusable Professional Website Architecture

Designed and built a working professional website architecture using independently testable content pages, thin Jekyll wrappers, reusable includes, responsive presentation, shared navigation, C-3PO document viewing, source viewing, sharing, and synchronized résumé publishing across HTML, PDF, DOCX, TXT, ODT, and Markdown. The architecture is intended to be reusable for other personal and professional websites.

EDUCATION

Oberlin College — Bachelor of Arts, Russian/Soviet Studies

Los Angeles City College — Computer Science studies; associate's degree nearly complete

University of Massachusetts Lowell — Approximately 1 year of undergraduate mathematics study

ACTIVE CREDENTIALS

- California Resident Insurance Producer License
- California Security Guard Card
- California Class C Driver's License
- California 30-Day Substitute Teaching Permit